

5ARE0, Academic Year 2026 – 2027

Assignment 1

Assessing Bicycle Lane Quality Using Smartphone Sensor Data

As cities around the world strive to become more bike-friendly, the quality of biking infrastructure has become a critical issue. Smooth and well-maintained biking lanes are essential for the safety and comfort of cyclists. However, many bike lanes suffer from poor maintenance, leading to uneven surfaces with potholes, cracks, bumps, and other imperfections. These issues not only make cycling uncomfortable but also increase the risk of accidents and injuries.

In urban environments, cyclists frequently encounter varying lane conditions that can impact their ride experience and safety. Identifying and addressing these problematic areas is crucial for urban planning and the maintenance of biking infrastructure. However, manual inspection of bike lanes can be time-consuming and resource intensive. As a result, there is a growing interest in developing automated tools to monitor the condition of bike lanes using technology that is readily available to cyclists.

Smartphones come equipped with sensors such as accelerometers, gyroscopes, and gravity sensors. These sensors can capture data related to the movement and orientation of the cyclist and their bike. By analyzing this sensor data, it is possible to detect when a bike lane is smooth or bumpy. Automated detection of lane conditions can provide valuable data for city planners and maintenance crews, allowing them to prioritize repairs and improve the overall biking experience.

1. Aim and Research Questions

The aim of this assignment is to gather and preprocess the data required to develop a tool that can automatically determine the condition of bike lanes—whether they are smooth or bumpy. The project will explore the use of sensor technology for data collection and apply basic techniques for cleaning and preprocessing the data. This process involves capturing sensor data from smartphones as cyclists ride over various lane surfaces.

2. Protocol

2.1 Participants & Activity Selection

Groups of 3 are assigned, in each group, 2-3 participants will be recruited.

Inclusion criteria:

- Participant knows how to bike.
- Participant is part of this course (5ARE0).
- In each group, all participants are member of the same assignment group.

Exclusion criteria:

- Any medical condition (including sports injuries) that does not allow biking.
- Does not know how to bike.

2.2 Measurements

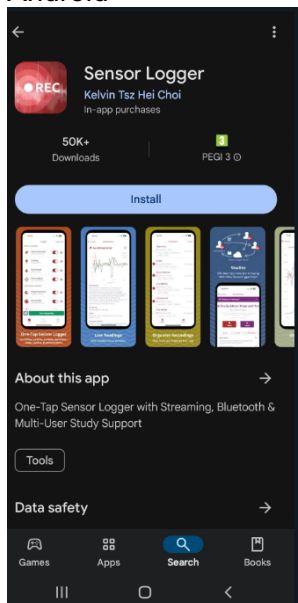
Use the free **Sensor Logger** app (iOS/Android) configured to record:

- **Accelerometer** (captures linear acceleration in 3D space)
- **Gyroscope** (captures angular velocity and rotational movements)
- **Gravity sensor** (provides device orientation relative to Earth's gravity)

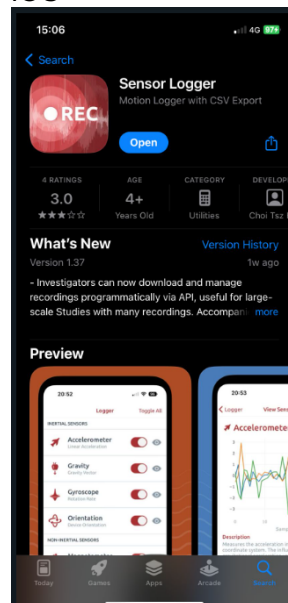
Keep all other sensors **OFF**. Use **default sampling rates of 100 HZ**. Ensure **Standardisation** option is activated to ensure consistency among different operating systems.

Tip: Secure the phone consistently—e.g., in a tight pocket, armband, or waistband—so that orientation is reproducible across trials and participants. Document your placement strategy as it will affect data interpretation. Justify your experimental design choices and hyperparameters.

Android



iOS



(For iPhone) If you have issues accessing the (live) calibrated data of the Sensor Logger app, you need to enable motion and compass calibration on your phone as explained [here](#). You will notice this if the live view and the export yield empty results.

2.3 Procedures

Pre-Data Collection:

1. **Safety Assessment:** Ensure all participants can safely perform each activity
2. **Environment Setup:** Use consistent environments
3. **Phone Placement:** Standardize device placement

Data Collection Protocol:

1. Setup Recording

- Launch Sensor Logger with specified sensor configuration
- Attach phone to designated body location consistently

2. Activity Performance

- When selecting a biking lane for data collection, a typical red cycling lane without imperfections is considered smooth.
- When selecting a bumpy biking lane, please choose a lane that exhibits rough or bumpy features but is still commonly used by cyclists. The lane should have some noticeable imperfections, such as small potholes, cracks, or uneven surfaces, but it should not be excessively dangerous or pose significant safety risks.
- The goal is to capture typical variations in lane quality that cyclists encounter during regular use, not extreme or hazardous conditions.
- Make sure the lane you choose is one that you would feel comfortable biking on regularly and that is representative of the kind of surfaces urban cyclists frequently navigate. Your safety is a priority, so avoid any lanes that are in poor repair to the extent that they could be considered unsafe.
- How to collect data: start recording with the sensor, put the phone in your pocket, ride the bike along the lane for a given time, stop biking, take out your phone and stop recording.
- The length of the recording, the number of recordings and locations are left for group members to agree upon, justify and document.

2.4 Data Management

A clear and consistent approach to data management ensures reproducibility and reliability. Students should systematically document their activities, maintain uniform sensor setups and trial conditions, and have all group members contribute data. The team should consolidate the datasets with clear labeling and timestamps. Raw and processed files should be stored separately, named consistently, and anonymized to protect privacy.

3. Analysis & Modeling

Your analysis must demonstrate a complete data science pipeline with the following components:

3.1 Data Preprocessing

Make use of the techniques learned during the course and instruction sessions.

Tip: visualizing your features helps guide your preprocessing.

3.2 Feature Engineering & Extraction

Consider windowing strategy for time series-analysis. Transform raw sensor signals into meaningful representations using techniques covered in the course. Apply feature extraction and feature selection techniques to identify the most discriminative features.

3.3 Supervised Learning Approach

Train four supervised models using the methods presented to you during this course, specifically from the beginning of the course up to the assignment's deadline.

3.4 Unsupervised Learning Approach

Train four unsupervised models using the methods presented to you during this course, specifically from the beginning of the course up to the assignment's deadline.

3.5 Model Comparison & Analysis

Supervised vs. Unsupervised Comparison:

- Compare the effectiveness of both approaches.
- Analyze advantages and limitations of each method.
- Discuss scenarios where each approach might be preferred.
- Identify one model as your optimal model of choice.
- Identify optimal features from implemented feature selection techniques.

Model Justification:

Provide reasoning grounded in:

- **Problem Suitability:** How well does the model match the characteristics of activity recognition?
- **Data Requirements:** Is the model appropriate for your dataset size and feature complexity?
- **Interpretability:** Can the results be understood and trusted by the end users?
- **Computational Efficiency:** Is the model practical for real-time or mobile deployment?
- **Performance:** How does accuracy compare across different models?

Evaluation Metrics:

- **Metrics:** Each group has the liberty of deciding which evaluation metrics to use, provided they align with the task at hand. Selecting the evaluation metrics is an exploratory task and not necessarily bounded by metrics provided in this course.
 - **Illustrations:** Include tables and visualizations (plots, confusion matrices, feature selection plots, preprocessing, EDA) to support your findings and make results as clear as possible.
 - **Deployment:** In addition to evaluating your own data, you will receive an independent external dataset. This dataset must not be used for training or testing. Instead, after finalizing your models and selecting the best-performing one, you will deploy it on the external dataset to assess bike-lane quality. You will then create a visualization highlighting which sections of the road are smooth and which are bumpy.
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4. Reporting

4.1 Single ZIP Submission

Create one ZIP archive named `groupX_submission.zip` containing:

1. **Jupyter Notebook** (`groupX.ipynb`)
2. **Data collection and analysis report** (`groupX_Report.pdf`)
3. **Data Folder** (`data/`) with organized raw sensor files
4. **README file** (`README.md`) with setup instructions and dataset description

4.2 Notebook Requirements:

- **Complete Pipeline:** The notebook must execute end-to-end, without external dependencies besides having to import the source data. Therefore, no `.py` files should be imported to the notebook, the notebook must contain all the codes in-house, supported by structured markdown cells giving titles and sub-titles to different project stages.
- **Size of the notebook:** size of the executed notebook can vary from 4 – 50 MB.
- **The notebook status:** the notebook must be delivered in its executed form. Having said that, it is expected of the instructors to re-run the notebook to validate the results and the outputs. Therefore, run the notebook before submission.
- **Notebook content** should ideally cover: **Data Loading, Preprocessing, feature engineering, feature selection, training, evaluation, model selection, deployment.** The quality of your notebook is significant to your grading.
- **Reproducibility:** Include Python version and required packages (with versions) at the top.
- **Code Quality:** Use clear structure, meaningful variable names, adequate comments, and section organization with sufficient markdown cells.
- **Documentation:** Provide clear explanations of methodology and rationale for each step
- **Do not use other python packages besides the ones used during the instruction sessions.**

4.3 Report requirements

Write your report based on CRISP-DM Framework:

Conclude your notebook with a comprehensive analysis (**up to 8 pages**) covering:

Business/Objective Understanding

- Problem definition and relevance to health monitoring

Data Understanding

- Dataset characteristics, collection methodology
- Exploratory data analysis and key insights
- Data quality assessment and limitations

Data Preparation

- Preprocessing pipeline and rationale.
- Feature engineering/extraction/selection strategy.

Modeling

- Unsupervised and supervised approaches implemented.
- Model selection rationale and hyperparameter tuning.
- Comparison methodology between approaches.

Evaluation

- Performance metrics and validation strategy.
- Model comparison results.
- Analysis of classification errors and failure modes.

Deployment Considerations & Recommendations

- Practical implications for real-world implementation.
- Limitations and potential improvements.
- Recommendations for future work and clinical applications.

4.4 Submission guidelines

- Upload **only** the `groupX_HAR_submission.zip` file to Canvas.
 - Ensure all group member names and student numbers appear on the notebook title page.
 - **Reproducibility Check:** Instructors will run your notebook to validate results. Any failure in code execution or missing dependencies may impact your grade.
 - This group assignment constitutes **20%** of your final grade and should be submitted by **October 4th, 20:00**.
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5. Ethical and Privacy Considerations

- All activities are performed by **consenting adult participants** with full understanding of the research purpose.
- Always ensure participant safety—stop any trial if discomfort or injury risk arises.
- **Data confidentiality:** All sensor data remains anonymous and is used exclusively for this educational assignment.
- **Informed participation:** All group members understand data collection procedures and analysis goals.
- **Data retention:** Raw data will be deleted after course completion and final grade assignment.

Good luck!